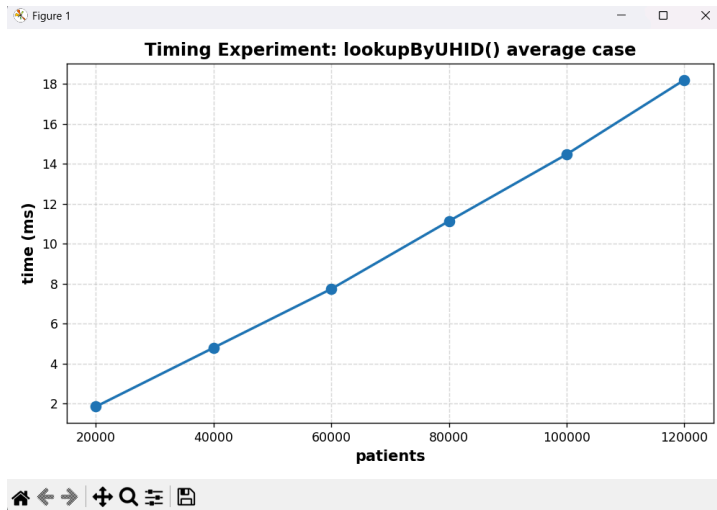


Assignment 1: Analysis document

1. My programming partner is Zewei Ren and I submitted the program to Gradescope.
2.
 - 2.1. I prefer the driver role in paired programming because I enjoy writing and modifying code according to my own ideas, and improving and simplifying code is a very rewarding thing for me.
 - 2.2. We applied pair programming techniques effectively in this assignment; I used these techniques to write about 40% of the code.
 - 2.3. In the next assignment, I hope we can have a brainstorming session before we officially start writing to provide multiple ideas for the assignment, so as to reduce the number of times the assignment code needs to be optimized.
 - 2.4. It took me about 4 hours to complete the programming and testing parts of this assignment.
 - 2.5. I felt that my programming partner and I communicated very frequently during the programming process, and I think he had a very positive attitude towards the assignment. I will continue to collaborate with him.
3. JavaThe difference between the Comparable and Comparator interfaces is that the Comparable interface can only use one sorting method at a time (usually natural ordering, such as sorting int from smallest to largest), while the Comparator interface can define multiple custom sorting methods (e.g., ranking by first name, then by last name, and finally by ID in an assignment). When using the Comparable interface, the elements themselves must implement the interface, and then the compareTo method is called using the element name as a dot notation. The Comparator interface, on the other hand, can be called through an object using the Comparator interface object name.compareTo comparison. Use the Comparable interface when comparing only one natural dimension of the sorted list, and the Comparator interface when comparing multiple components. In the assignment, you can...ListType implementationThis can be accomplished using the methods of the Comparable interface, changing the generic type <ListType> to...<ListType> extends Comparable<ListType>> Then change the compareTo method to the compare method.



4.

```
problemSizeDescription = "patients"  
problemSizeMin = 20000  
problemSizeCount = 6  
problemSizeStep = 20000
```

5. Execution time increases linearly with size. This is consistent with the timing behavior I observed in lookupByUHID within the Facility.